UNIT – 4

ASSESSMENT – 1

REPORTS

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Title 1:

Data Preparation and Inductive Learning for Diabetes Prediction

Aim

To prepare patient healthcare data for machine learning and develop a suitable inductive learning framework for diabetes prediction.

Methodology

Patient information is divided into input features and a target variable. The dataset is divided into 70% training and 30% testing data using stratification. Numerical features are standardized and class imbalance is handled using class weights.

Result

The patient dataset was successfully prepared for machine learning. The data was divided into training and testing subsets and numerical features were standardized.

Conclusion

Proper data preparation improves the reliability of subsequent diabetes prediction models. Class weighting helps the model pay greater attention to diabetic cases.

Title 2:

Decision Tree Based Diabetes Diagnosis

Aim

To develop and evaluate a Decision Tree classifier for predicting diabetes risk.

Algorithm

1. Load patient dataset.
2. Separate features and target.
3. Split data into training and testing sets.
4. Build Decision Tree using Gini Index.
5. Train the model.
6. Predict test cases.
7. Calculate Accuracy, Precision, Recall and F1-score.
8. Identify False Negatives.
9. Apply cost-complexity pruning.
10. Compare the models.

Result

The Decision Tree successfully classified patients into diabetic and non-diabetic categories. The confusion matrix identified False Negatives, which are especially important for diabetes screening.

Conclusion

A Decision Tree provides an interpretable model for healthcare classification. Pruning can reduce unnecessary complexity and overfitting while maintaining useful predictive performance.

Title 3:

Statistical Learning for Diabetes Risk Stratification

Aim

To apply Logistic Regression for diabetes risk prediction and compare its performance and feature importance with the Decision Tree model.

Method

1. Preprocess the dataset.
2. Standardize numerical features.
3. Split into training and testing data.
4. Train Logistic Regression.
5. Calculate Accuracy and AUC-ROC.
6. Extract model coefficients.
7. Rank the features according to coefficient magnitude.
8. Compare the top predictors with the Decision Tree.

Result

Logistic Regression successfully classified the diabetes cases and provided probability-based predictions. Glucose, BMI and age are expected to be among the most influential predictors.

Conclusion

Statistical learning provides a useful baseline because it is relatively simple and interpretable. Logistic Regression can be preferable when probability estimates and coefficient-based interpretation are important.

Title 4:

Q-Learning Based Personalised Treatment Recommendation

Aim:

To model personalised healthcare action selection as a Markov Decision Process and train a Q-Learning agent to learn an action policy.

MDP Components

Component	Definition
States	Low, Moderate, High and Critical Risk
Actions	Diet, Exercise, Medication, Monitor
Reward	Health-state improvement
Transition	Change from one health state to another
Policy	Action with highest Q-value

Algorithm:

1. Initialize Q-table with zeros.
2. Select an initial patient state.
3. Select an action using ε-greedy strategy.
4. Observe the next state.
5. Calculate reward.
6. Update Q-value.
7. Repeat for multiple steps.

8. Repeat for at least five episodes.
9. Select the action with maximum Q-value.
10. Generate the final policy.

Q-Learning Formula

Result

The Q-Learning agent learns Q-values for different health states and actions. The action with the highest Q-value becomes the learned policy for each state.

Ethical Considerations

- Patient safety must be prioritized.
- The model must not make unsupervised medical decisions.
- Bias in training data must be evaluated.
- Recommendations should be explainable.
- Patient privacy must be protected.
- Human/clinical supervision is required.

Conclusion

Q-Learning demonstrates how reinforcement learning can model sequential decision-making for healthcare. However, the simplified environment used in this academic implementation cannot be directly used for real patient treatment. A real system would require clinical validation, safety constraints, ethical review and extensive testing.